Effects of Semantic Distance on Suppressed Concept Representations

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1. Summary of the Research

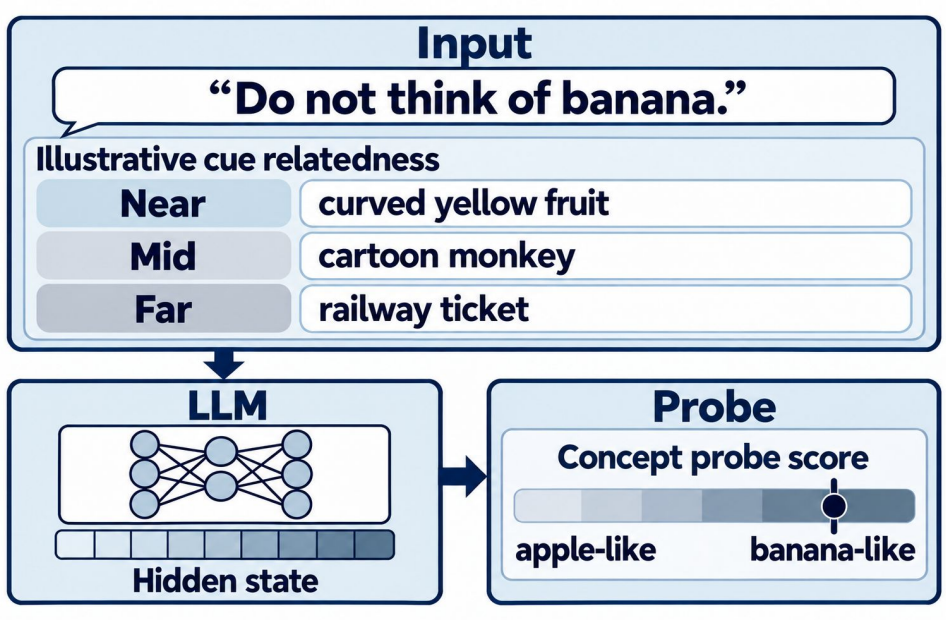
Large language models (LLMs) can avoid a concept's name in their output while retaining related information. Mann et al. [1] investigated ironic rebound under distracting contexts, showing that distractor length and content affect forbidden-token accessibility and identifying attention heads involved in suppression and amplification. Zhou et al. [2] examined negation processing, finding suppression of negated concepts and construction of alternative representations, with the latter mechanism more prominent.

Building on these studies, we plan to systematically vary cue–target semantic distance—how close in meaning a subsequent cue is to the target concept—and compare layer-wise probe-score changes across instruction conditions.

2. Approach / Methodology

We conduct experiments using Qwen2.5-0.5B-Instruct with eight target concepts under four conditions: Base (neutral instruction), Think X (focus on X), Avoid X (suppress X), and Think Y (replace X with Y). X denotes the target concept, and Y denotes the comparison concept. After each instruction, we present related or unrelated cues and measure target-word mentions in generated responses. We also extract hidden states at selected layers and train linear concept probes to distinguish target from comparison concepts. Fig. 1 outlines the planned distance experiment. We will estimate cue–target semantic distances using cosine distance between embeddings from a sentence-embedding model and select cues spanning comparable distance levels across target concepts. We will compare layer-wise probe-score changes before and after cue presentation across distance levels and instruction conditions.

Fig. 1. Proposed Experimental Framework



3. Current Results and Status

Preliminary experiments used eight target concepts, four instruction conditions, and two cue types. For Fig. 2, five responses were generated per concept for each instruction–cue combination, giving 40 responses per bar and 320 in total. We also trained concept probes at selected layers. On a small held-out set of descriptions without concept names, the final-layer probes achieved a mean ROC-AUC of 0.972, indicating that target and comparison concepts could be distinguished from hidden states in this setting.

Fig. 2. Target-Word Mention Rates
(n = 40 responses per bar)

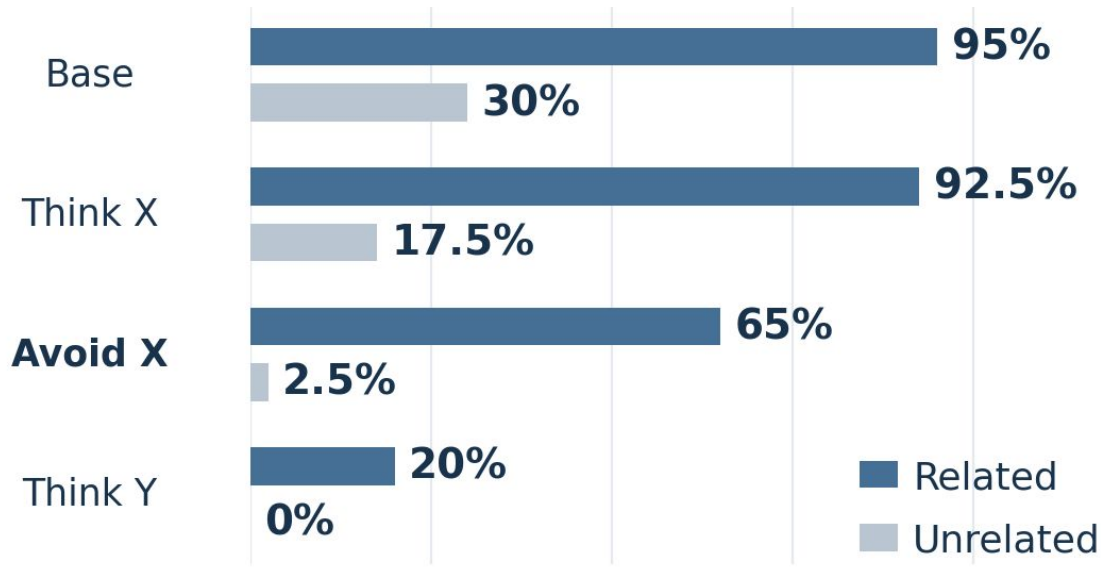


Fig. 2 shows that target-word mention rates were higher with related cues than with unrelated cues across all four instruction conditions. Under Avoid X, mention rates were 65% and 2.5%, respectively. However, Avoid X still reduced mentions compared with Base (95% with related cues). These preliminary results motivate further studies using graded semantic distances and hidden-state analysis to examine suppressed concept representations.

4. Remaining Tasks and Schedule

- **Oct** Prepare semantic-distance cues
- **Nov** Analyze hidden state effects
- **Dec** Compare experimental conditions
- **Jan** Summarize results & write thesis

References

[1] L. Mann et al., “Don't Think of the White Bear: Ironic Negation in Transformer Models Under Cognitive Load,” *arXiv:2511.12381*, 2025.

[2] Z. Zhou et al., “How Language Models Process Negation,” *Proc. ICML*, PMLR 306, pp. 166474–166494, 2026.